

75 participants
38 controls · 37 PD
597 task recordings

Participant-stratified folds
5 outer folds × 5 repeats
Shared across tasks / methods

Outer test participants
Held out during tuning
and model training

Outer training participants
Classical tuning: 3 inner folds
CNN epoch count: first inner fold

LPQ-RGB: 64×64 input
768 histogram bins
SVM · LR · random forest

CNN: 128×128 input
4 convolutional blocks
105,873 parameters

External predictions
Eight task-level scores
Participant task fusion

Paired inference
Participant bootstrap across tasks / repeats
Paired permutation tests · Holm correction

CNN explanations on frozen test models
Grad-CAM · occlusion · channel removal
Deletion · weight-randomization checks